recurrence, where everything that has happened will happen again, in exactly the same way, forever.[fn2] In fact, both these seemingly opposing theories are commonly associated with Friedrich Nietzsche, who, after proclaiming the death of any god that might bring order and meaning to the universe, grappled with the implications of living in a cosmos lacking a final redemption arc.

Nietzsche isn't the only one to have contemplated the meaning of existence, of course. Everyone from Aristotle to Lao-Tzu to de Beauvoir to Captain Kirk to Buffy the Vampire Slayer has at one point asked, "What does it all mean?" As of this writing, we have yet to reach a consensus.

Whether or not we subscribe to any particular religion or philosophy, it would be hard to deny that knowing our cosmic destiny must have some impact on how we think about our existence, or even how we live our lives. If we want to know whether what we do here ultimately matters, the first thing we ask is: how will it come out in the end? If we find the answer to that question, it leads immediately to the next: what does this mean for us now? Do we still have to take the trash out next Tuesday if the universe is going to die someday?

I've done my own scouring of theological and philosophical texts, and while I learned many fascinating things from my studies, unfortunately the meaning of existence wasn't one of them. I may just not have been cut out for it. The questions and answers that have always drawn me in most strongly are the ones that can be answered with scientific observation, mathematics, and physical evidence. As appealing as it sometimes seemed to have the whole story and meaning of life written down for me once and for all in a book, I knew I would only ever really be able to accept the kind of truth I could rederive mathematically.

LOOKING UP

Over the millennia since humanity's first ponderings of its mortality, the philosophical implications of the question haven't changed, but the tools we have to answer it have. Today, the question of the future and ultimate fate of all reality is a solidly scientific one, with the answer tantalizingly within reach. It hasn't always been so. In Robert Frost's time, debates still raged in astronomy about whether the universe might be in a steady state, existing unchanging forever. It was an appealing idea, that our cosmic home might